الشبكات الوهمية والخزانات الخفية

Lecture 4 — Virtual Networks and Virtual Storage

Q/ Why must the network be built in software in the cloud ?

1. **Isolation per tenant**: each customer needs a private network others can't see, even on the same physical switches.
2. **Agility**: a network, subnet or firewall rule is created in seconds by an **API call**, not by a technician with a cable.
3. **Mobility**: when a VM live-migrates it keeps its IP and network policy — the network must **follow the VM**.

Virtual NICs and switches

> **Virtual NIC (vNIC)** : a software network card given to a VM, with its own MAC address.

> **Virtual switch (vSwitch)** : forwards frames between VMs on the same host **in software**; frames for other hosts leave through the physical uplink.

Examples: VMware vSwitch, Hyper-V vSwitch, Linux bridge, **Open vSwitch (OVS)**, Docker bridge.
Security: traffic between two VMs on one vSwitch **never meets the physical firewall** — use port security and limit promiscuous mode.

Q/ Compare the VM network modes.

Mode	How the VM connects	Reaches the internet	Reachable from the LAN	Typical use
Bridged	A separate device on your real network with its own LAN address	Yes	Yes	Servers other machines must reach
NAT (Lab 3)	Shares the host's address; VMware translates	Yes	No (unless port forwarding)	Labs, safe default
Host-only	Private network between VM and host only	No	No	Isolated testing, malware analysis

Rule: choose the **least exposed** mode that works. Bridged puts the VM directly on your LAN.

VLAN

> **VLAN (Virtual LAN)** : splits one physical switch into several separate **broadcast domains**.

Q/ How does a VLAN work ?

1. **Access ports** belong to one VLAN.
2. **Trunk ports** carry several VLANs; each frame is tagged with **802.1Q (4 bytes)**.
3. The VLAN ID is **12 bits → 1 to 4094 usable**.
4. Different VLANs talk only through a **router or Layer 3 switch**.

Q/ Why are VLANs not enough for the cloud ?

1. Only **4094** IDs — a cloud has thousands of tenants.
2. Each VLAN is a flat **Layer 2** domain tied to one switched network.

Underlay and overlay

> **Underlay** : the real, physical routed IP network (e.g. leaf-spine).

> **Overlay** : the virtual network the tenant sees, built **on top of** the underlay with tunnels.

Like a letter in an envelope: the postal network reads only the **outer** address. Two tenants can even reuse the same 10.0.0.0/24 without conflict.

Q/ What are the four steps of an overlay (VXLAN) ?

1. **Send**: VM A sends a normal frame to VM B (it doesn't know a tunnel exists).
2. **Wrap**: the VTEP on Host 1 looks up (VNI, MAC B), finds Host 2, and adds outer Ethernet + IP + **UDP 4789** + VXLAN headers (~50 bytes).
3. **Route**: the underlay routes by the **outer IP only**; tenant addresses stay hidden.
4. **Unwrap**: the VTEP on Host 2 reads the VNI, removes the outer headers, and delivers the original frame to VM B.

> **VXLAN** : wraps the tenant's whole Ethernet frame inside a **UDP** packet — "Layer 2 over Layer 3".

> **VNI (VXLAN Network Identifier)** : 24 bits → about **16 million** virtual networks (2^{24}).

> **VTEP (VXLAN Tunnel Endpoint)** : sits in the host's vSwitch or a leaf switch, and wraps/unwraps frames.

Q/ What is the VXLAN header layout ?

- Outer Ethernet **14 B** · Outer IP **20 B** · UDP **8 B (port 4789)** · VXLAN header **8 B (carries the VNI)** · inner Ethernet frame → ≈ **50 B overhead**.

> **MTU pitfall** : the 50 extra bytes can push packets over 1500 → raise the underlay MTU (e.g. **9000**) or lower the VM MTU to about **1450**.

Security: **VXLAN does not encrypt or authenticate**. The VNI separates tenants only logically — protect the underlay and use IPsec or WireGuard for sensitive traffic.

Aspect	VLAN	VXLAN
Segment ID	12 bits: 4094 usable	24-bit VNI: ~16 million
How it isolates	A tag on the Ethernet frame (802.1Q)	The frame is wrapped in UDP/IP (tunnel)
Reach	Within one switched Layer 2 network	Across a routed IP network, even between sites
Overhead	4 bytes	~50 bytes
Configuration	Switch ports and trunks, often manual	Tunnel endpoints + control plane (e.g. BGP EVPN), automated
Best for	Campus, small or fixed data centers	Multi-tenant clouds, large data centers

In practice: VLANs inside a rack/campus, VXLAN across the data-center fabric.

Q/ Map the public-cloud network building blocks.

Building block	Does	AWS	Azure
Virtual network	Your private address space	VPC	VNet

Building block	Does	AWS	Azure
Subnet	A slice, public or private	Subnet	Subnet
Route table	Where traffic goes next	Route table	Route table (UDR)
Internet access	In/outbound paths	Internet gateway, NAT gateway	Public IP, NAT Gateway
Firewall rules	Allow/deny by port, protocol, source	Security group (**stateful**), NACL (**stateless**)	NSG

Default deny. **Never** allow SSH (22) or RDP (3389) from **0.0.0.0/0**. Keep databases in **private** subnets.

SDN

> **SDN (Software-Defined Networking)** : separates the **control plane** (decides the rules, centralized in a controller) from the **data plane** (switches that only forward packets by the rules).

Traditional network	SDN
Every switch runs its own control logic, configured one by one	Logic centralized in software (the controller)
Hard to change and automate	One policy change reaches every switch in seconds, via API

Q/ What are the SDN layers and APIs ?

1. **Applications** (routing, firewall, load balancing, monitoring) — talk to the controller through the **Northbound API (REST)**.
2. **SDN controller** = control plane, global view.
3. **Switches** = data plane — controlled through the **Southbound API (OpenFlow)**.

> **OpenFlow** : a southbound protocol; the switch keeps **flow tables** — if a packet **matches** these fields, perform this **action**.

Priority	Match	Action
300	src 10.0.1.5, TCP 22	Drop
200	dst 10.0.2.0/24	Forward to port 3
100	VLAN 20	Forward to port 5
0	anything else (**table-miss**)	Send to controller

Q/ What happens when no flow entry matches ?

1. The switch sends a **packet-in** message to the controller.
2. The controller decides and replies with a **flow-mod**.
3. Later packets of that flow match the new entry and are forwarded at full speed.

OVS is the most common OpenFlow software switch; **P4** goes further and programs how the switch parses packets.
Security: the controller is a **high-value target** — whoever controls it controls the network. Isolate it, authenticate switches, use TLS, restrict API access.

Storage

	Block	File	Object
Data unit	Fixed-size blocks	Files in folders	Object + metadata + ID
Access	Raw disk: iSCSI, Fibre Channel	NFS, SMB over the network	HTTP **REST API**
Examples	AWS EBS, Azure Disk, SAN	AWS EFS, Azure Files, NAS	AWS **S3**, Azure Blob, MinIO
Strength	Low latency, high speed	Familiar, shared by many users	Huge scale, cheap, durable
Limit	One VM at a time, no sharing	Scaling and locking get harder	Not for databases or editing in place
Best for	VM disks, databases	Shared folders, home directories	Backups, media, logs, big data

Traditional storage	Software-defined storage
DAS (disks in one server), NAS (file server), SAN (dedicated block network)	Software on commodity servers
Scale-up: buy a bigger, costlier array	**Scale-out**: add nodes, capacity and speed grow
Vendor hardware and tools	Pools many disks into one logical store; policies by API
	Examples: **Ceph, HDFS, VMware vSAN, MinIO**

Replication : 3 full copies on different nodes — simple and fast, but **3× the space**.

Erasure coding : data split into 4 pieces + 2 parity — survives **2 failures at 1.5× space**.

Replication protects against **failure**, not against **deletion or ransomware**: keep separate backups.

Security lens

Area	Typical risk	Good practice
VLAN & vSwitch	VLAN hopping via trunk misconfiguration; unseen east-west traffic	Disable unused ports, fix native VLAN, micro-segmentation
Overlay (VXLAN)	Unencrypted tunnels; attacker on the underlay reads/injects	Secure the underlay; IPsec or WireGuard
Cloud network rules	Security groups open to 0.0.0.0/0; exposed databases	Least privilege, private subnets, rule audits
SDN controller	Single point of control and failure; rogue switches	Isolate, TLS, authenticate, redundant controllers
Storage	Public buckets, unencrypted data, leaked snapshots/backups	Block public access, encrypt at rest & in transit, manage keys, versioning

Q/ Mini case — university private cloud (Teaching, Research, Administration): which storage for each need ?

1. Exam PDFs for 2,000 students → **object** (scale, cheap, read-mostly).
2. Lab VM disks → **block** (low latency, one VM per disk).
3. Shared Linux home folder → **file** (NFS/SMB, shared).

- Keep student VMs away from admin: separate segments (VLAN/VXLAN per tenant) + default-deny firewall rules. VLANs may be enough for 3 tenants; move to **VXLAN** when tenants/segments grow or span sites.

Summary of Cloud Computing L4: Virtual Networks & Virtual Storage. Built only from the lecture file; check the original slides for anything you're unsure about.